

* Type conversion in java

Implicit

→ automatically
done by java

Explicit

→ we have to
do it manually

byte b = 24;

int i;

i = b;

System.out.println(i); // 24

* Rule for implicit conversion

→ Destination data type should be wider than source data type
(widening conversion)

Explicit Conversion (Casting)

int i = 300; → 100101100_{LSB}

byte b;

✗ b = i; // Cannot convert int to byte

✓ b = (byte) i;

↳ casting
i to b

System.out.println(b); // 44

byte
~~0000~~ 00101100_{8 bit}

or

b = 300 % (Range of byte)

b = 300 % (256)

b = 44

long → int

double → float

↳ narrowing
conversion

* Truncating Conversion

float
or
double $\rightarrow$ int

float $f = 16.25$ f;

int i;

$i = f;$ // Error

$i = (int) f$ // $i = 16$

* Automatic Type promotion

byte a = 50;

byte b = 40;

byte c = 100;

int i = (a * b) / c;

50×40
 $= 2000$

(out of range)

-128 to +127

byte $\rightarrow$ int

$\therefore$ byte promoted to become an integer

byte b = 50;

b = b * 2; X

50×2
(int)

= 100 (int)

type mismatch
casting req

b = (byte) b * 2;